

SHUBHAM KISHOR KALE

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AI Engineer and Data Scientist building agentic RAG/LLM systems, production ML with measured impact, and large-scale ELT pipelines.

EDUCATION

Stony Brook University, New York, USA | *Master of Science in Data Science* | **GPA: 3.78/4** **Aug 2024 - Dec 2025**
NMIMS University, Mumbai, India | *Bachelor of Technology in Data Science* | **CGPA: 3.56/4** **Jul 2019 - May 2023**

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, SQL, JavaScript, TypeScript, Bash

Concepts & Domains: AI Engineering, Data Engineering, Data Science, Machine Learning, Deep Learning, NLP, Generative AI (RAG, LLM Agents, MCP, Fine-tuning), Statistical Analysis, Causal Inference, A/B Testing, Feature Engineering, Dimensional Modeling, Time Series Analysis, Data Structures & Algorithms, System Design, REST APIs, Server-Sent Events (SSE), OOP & Design Patterns, ETL, Data Ingestion, Data Warehousing, Data Modeling, Data Security, Distributed Systems, Finance, Healthcare, Insurance

Frameworks, Tools & Cloud: PyTorch, scikit-learn, XGBoost, Hugging Face, Claude API, Gemini API, LangChain, LangGraph, Pandas, NumPy, Kafka, Docker, Apache Airflow, Dagster, dbt, AWS (S3, Redshift, IAM), GCP (BigQuery), Snowflake, PostgreSQL, FastAPI, Streamlit, React, Next.js, Pydantic, pytest, Git, GitHub Actions (CI/CD), Tableau, Power BI

WORK EXPERIENCE

Quantegy Analytics | New York, USA **May 2025 - Present**
Analytics Engineer

- Shipped an LLM-powered navigation assistant (Claude API) into a production analytics platform serving 50+ daily users across 5 markets, routing natural-language queries to the right dashboard via a structured page-catalog context.
- Ranked retail endcap placement across 420 stores for a consumer products client with a stacked decision-tree and linear-regression ensemble over engineered demographic features; installed stores posted a 12% first-week sales lift (\$118K) versus 0.2% in matched controls.
- Delivered the underlying Streamlit analytics platform for a healthcare client, cutting a 3-day Excel reporting cycle to same-day.
- Engineered Dagster-orchestrated ELT pipelines ingesting 4M+ patient and financial records from 15+ Redshift tables into Snowflake, powering analytics across 3 healthcare markets.
- Modeled the target Snowflake schemas behind 3 client data platforms: fact and dimension tables with conformed dimensions, declared grain, and Type 2 history tracking across raw vendor EHR, API, and CSV feeds.
- Owned data engineering and delivery for a correctional healthcare client: Dagster pipelines ingesting 120+ vendor EHR and CSV tables into Snowflake across 15 counties, plus a production Streamlit app with per-county, per-role access control.
- Accelerated multi-table ingestion ~5x with parallelization, REST APIs, S3 archival, and Pydantic validation in a Dockerized CLI adopted by 4+ teams; scheduled on Airflow with pytest/httpx validating 10M+ records across 10+ tables.

Mettler-Toledo | Mumbai, India **Mar 2024 - Jun 2024**
Data Analyst Intern

- Designed an end-to-end RAG system with fine-tuned T5/BART models (Hugging Face) for automated KPI commentary, with vector search over 10M+ financial records (85%+ relevance); removed 15 hrs/month of manual write-ups.

Marcellus Investment Managers | Mumbai, India **Mar 2023 - Nov 2023**
Quantitative Analyst Intern

- Administered the PostgreSQL research environment end to end (backups, recovery, partitioning, role-based access, query and index tuning) and consolidated 25 tables (3M+ rows) across 2 schemas with Python automation that cut 10 hrs/week of reconciliation.
- Developed a regression-based impact-cost model ($R^2 = 0.82$) for trade-execution optimization and a forensic-screening ML ensemble (84% accuracy, 87% specificity) adopted into ongoing fraud-risk screening.

PROJECTS

AI Financial Research Agent · [GitHub] · [Live Demo] · *Full-Stack GenAI, Agentic RAG, MCP, SSE, Evaluation*

- Architected an agentic RAG system over SEC 10-K filings (5 companies, 67K+ chunks) with LangChain, LangGraph ReAct agents, a FastMCP server, ChromaDB, and Gemini 2.5 Flash Lite, with tools discovered at runtime via MCP.
- Built and deployed a Next.js and TypeScript chat UI streaming source-grounded answers with live citations via Server-Sent Events over a two-service FastAPI and MCP backend; GitHub Actions CI/CD runs lint, eval, and pytest.
- Instrumented a RAGAS evaluation harness with a separate judge model to avoid quota contention with the agent's generation model; scored 0.80 faithfulness and 0.80 context recall across question types.

Grid Resilience: Real-Time Anomaly Detection · [GitHub] · *Kafka Streaming, Distributed Systems, ML Evaluation*

- Streamed 89K+ hourly NYISO load records across 11 zones through Apache Kafka (Docker/KRaft) with a stateful per-zone-hour join and adaptive rolling z-score anomaly detection; kappa 0.52, recall 0.61 versus an offline baseline.